Replication

Purpose

This document explains how to reproduce the main experimental results reported for the coursework submission.

Assumptions

1. The repository has been cloned locally
2. Commands are run from the repository root
3. Dependencies have been installed with `python -m pip install -r requirements.txt`

Replication Steps

Run the following commands in this exact order.

1. Regenerate repeated experiment results

Run:

```
python -m scripts.run_experiments
```

This produces:

results/experiments/raw_results.csv
results/experiments/aggregated_results.csv

The current experiment configuration uses:

- Algorithms: baseline_random_search and bestconfig
- Budgets: 50 and 100
- Runs per setting: 30
- Base seed: 42

2. Regenerate comparison plots

Run:

```
python -m scripts.make_comparison_plots
```

This produces:

- results/experiments/plots/boxplots/*.png
- results/experiments/plots/median_comparisons/*.png

3. Regenerate statistical test results
Run:
```
python -m scripts.run_stat_tests
```

This produces:
- results/experiments/stat_tests/wilcoxon_results.csv

The script applies:
- Paired Wilcoxon signed-rank tests
- Holm correction across the dataset-budget comparisons

4. Regenerate the final report table
Run:
```
python -m scripts.build_results_table
```

This produces:
- results/experiments/report_tables/final_results_table.csv
- results/experiments/report_tables/final_results_table.md

Reported Coursework Artefacts
The main artefacts used to support the coursework results are:
- results/experiments/raw_results.csv
- results/experiments/aggregated_results.csv
- results/experiments/plots/boxplots/*.png
- results/experiments/plots/median_comparisons/*.png
- results/experiments/stat_tests/wilcoxon_results.csv
- results/experiments/report_tables/final_results_table.csv
- results/experiments/report_tables/final_results_table.md

Clean Reproduction Sequence

```
1. python -m pip install -r requirements.txt
2. python -m scripts.run_experiments
3. python -m scripts.make_comparison_plots
4. python -m scripts.run_stat_tests
5. python -m scripts.build_results_table
```

This sequence reproduces the experiment outputs, plots, statistical test results, and final summary table reported in the coursework.